

Lesson 30: Discrete Random Variables — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. What must all probabilities in a distribution sum to?
2. P : 0.2, 0.3, k . Find k .
3. P : 0.1, k , 0.5, 0.2. Find k .
4. Can a probability be 1.3?
5. $E(X)$ for $x = 2, 4$ with $P = 0.5, 0.5$?
6. P : 0.25, 0.25, k . Find k .

Section 2 — Core practice

7. x : 0,1,2 with P : 0.5,0.3,0.2. Find $E(X)$.
8. x : 1,2,3 with P : 0.2,0.5,0.3. Find $E(X)$.
9. x : 1,2,3,4 with P : 0.1,0.4,0.3,0.2. Find $E(X)$.
10. x : 0,10 with P : 0.9,0.1. Find $E(X)$.
11. x : 5,10,15 with P : 0.2,0.5,0.3. Find $E(X)$.

12. x : 2,4,6 with P : 0.5,0.25,0.25. Find $E(X)$.

Section 3 — Challenge & exam-style

13. x : 1,2,3,4, P : 0.1,0.4,0.3,0.2. Verify $\sum P = 1$ and find $E(X)$.

14. A game: win £5 with prob 0.2, lose £1 with prob 0.8. Find the expected gain.

15. x : 0,1,2,3 with P : k , 0.3, 0.3, 0.1. Find k and then $E(X)$.

16. Is $E(X)$ always one of the possible values of X ? Explain with the fair die.